Cambridge O Level Computer Science

Chapter 5: The internet and its uses

Lecture notes with Check-Yourself questions and a full answer key.

In this chapter you will learn about:

- The internet and the World Wide Web: the differences between them, URLs, HTTP and HTTPS, web browsers, how web pages are located and retrieved, and cookies (session and persistent).
- Digital currency: digital currencies, cryptocurrency, and the process of blockchaining used to track transactions.
- Cyber security: threats (brute force attacks, data interception, DDoS, hacking, malware, phishing, pharming, social engineering) and solutions that keep data safe (access levels, anti-malware, authentication, firewalls, proxy servers, privacy settings, SSL).

The internet is one of the greatest inventions of the twentieth century, but it has also created new kinds of crime. This chapter looks at how the internet works and how to recognise and resist cyber security threats.

5.1 The internet and the World Wide Web

5.1.1 The internet vs the World Wide Web

The internet (INTERconnected NETwork) is a worldwide collection of interconnected networks: a concept rather than something tangible, relying on physical infrastructure. The World Wide Web (WWW) is only the part of the internet accessed with web browser software: a massive collection of multimedia web pages based on hypertext transfer protocol. The Web is a way of accessing information using the internet; they are quite different things.

Summary of differences:

Internet	**World Wide Web**
Worldwide collection of interconnected networks and devices	Collection of multimedia web pages and other information on websites
Users can send and receive emails	Resources accessed by web browsers
Allows online chatting (text, audio, video)	http(s) protocols; pages written in HTML
Uses TCP and IP transmission protocols	URLs specify the location of web pages

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Check yourself

1. 1. Explain the difference between the internet and the World Wide Web.
2. 2. Classify each statement as internet or WWW: (a) uses TCP and IP protocols, (b) pages are written in HTML, (c) allows online chatting, (d) URLs specify the location of web pages.

5.1.2 Uniform resource locators (URLs)

A URL is a text address for a website, typed into the browser address bar in this format: protocol://website address/path/file name

- Protocol: usually http or https.
- Website address: domain host (www), domain name (website name), domain type (.com, .org, .net, .gov) and sometimes a country code (.uk, .de, .cy).
- Path: the directory of the website (often omitted, giving the root page).
- File name: the item on the web page, for example ict.

Worked example: Identify the parts of this URL: https://www.schoolportal.ac.bd/students/timetable.html

Protocol: https (http with security).

Domain host: www. Domain name: schoolportal. Domain type: .ac. Country code: .bd.

Path (directory): /students.

File name: timetable.html.

Check yourself

3. 1. Write down the general format of a URL.
4. 2. For the URL https://www.shoponline.co.uk/books/bestsellers.html, identify the protocol, domain name, domain type, country code, path and file name.
5. 3. Explain the difference between a domain type and a country code, giving an example of each.

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5.1.3 HTTP and HTTPS

Hypertext transfer protocol (http) is the set of rules for transferring files across the internet. When security (SSL or TLS) is added, it becomes https: the 's' stands for secure. You will often see a green padlock in the status bar too. HTTPS is a more secure way of sending and receiving data across a network.

Check yourself

6. 1. What does the 's' in https stand for, and what does it indicate?
7. 2. Give two visual clues in a browser that a website is using https.

5.1.4 Web browsers

A web browser is software that accesses and displays web pages on your screen. It interprets (translates) the HTML sent from websites and shows the result: text, images, video and audio. Typical browser features:

- A home page; bookmarks (stored favourite websites); browsing history; forward and back navigation through opened pages.
- Multiple tabs so many pages can be open at once.
- Use of cookies; use of hyperlinks (click a link to open in the same tab, or ctrl+click to open in a new tab).
- A cache (storing page data for faster reloading); use of JavaScript; an address bar.

Check yourself

8. 1. Explain what a web browser does with the HTML sent by a website.
9. 2. Give four features of a web browser.
10. 3. What is the difference between clicking a hyperlink and ctrl+clicking it?

5.1.5 Retrieval and location of web pages

Websites are written in HTML and hosted on a web server that has its own IP address. Your browser needs that IP address, but users type URLs, not numbers. The Domain Name Server (DNS) system finds the IP address for a domain name: its servers hold databases of URLs matched to IP addresses, so users never have to memorise them.

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Locating a web page (example: www.hoddereducation.co.uk maps to 107.162.140.19):

11. The user types the URL; the browser asks DNS server (1) for the website's IP address.
12. If DNS server (1) cannot find it in its database or cache, it asks DNS server (2).
13. DNS server (2) finds the URL and maps it to the IP address; the address is sent back to DNS server (1), which stores it in its cache/database.
14. The IP address is sent back to the user's computer.
15. The computer contacts the website server; HTML files are downloaded and the browser interprets and displays them.

Check yourself

16. 1. What is the purpose of a DNS server?
17. 2. Describe the steps that occur when a browser cannot find a URL's IP address on the first DNS server it asks.
18. 3. Explain why URLs and DNS mean users do not need to memorise IP addresses.

5.1.6 Cookies

Cookies are small files (effectively look-up tables of key-data pairs, for example (surname, Jones), (music, rock)) sent by a web server and stored on the user's computer. When the user revisits, the browser reads the cookie, which holds preferences such as language, currency and browsing activity. Cookies track users, remember preferences and let sites customise pages (for example, 'Customers who bought Hodder IGCSE ICT also bought Hodder IGCSE Computer Science').

Session cookies vs persistent cookies

Session cookies	Persistent cookies
No expiry date; stored in temporary memory	Stored on the hard drive until the expiry date or until deleted
Deleted when the browser closes or the session ends	Remain after the browser closes or the session ends
Collect no information; do not personally identify the user	Remember login details and personal preferences
Example: keeping items in a virtual shopping basket	Example: automatic login without retyping details

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